



**UTTARANCHAL
UNIVERSITY
DEHRADUN**

**NAAC
GRADE A+**

Introductory Lecture

Lecture - Zero

Name of Faculty: Mr. Arpit Goel

Location of Faculty: Uttaranchal University

School Name: FUIT



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About the Course

- Course Name: COMPILER DESIGN
- Course Code: TCS-352
- L-T-P Structure: 3-1-0
- Weekly Engagement: 3
- Course objectives:
 - The prime objective of this course is to understand the principles and techniques used to perform translation and the fundamental concepts of compiler construction. Students will also Learn the process of translating a modern high-level language to executable code required for compiler construction.



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Mission of School

- To provide learning friendly ambience, developing competence in diversified areas creating excellence in technical arena.
- To establish worldwide knowledge integration platforms by developing intellectual capital through research activities.
- To establish networks and linkages with industries and academic institutes of National and International importance.
- To ensure optimum use of resource through infrastructure sharing, industry interface, faculty and students exchange programs.
- To produce ethically and morally strong work – force who would contribute to the development of knowledge economy.



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Vision of School

- To provide quality education, research, training and practical expertise that is stimulating and responsive to the needs of 21st century.
- To prepare students to excel in their profession by providing life long learning skills, sound theoretical knowledge, practical experience and all round development.
- To stress the importance of a vital sense of moral and ethical values and character building.
- To be at the zenith among leading Engineering Institutes in the coming decade and produce technical manpower of international standard and have an all pervasive environment of goodwill and credibility in the society.



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Graduation Attributes





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Program Outcomes (Pos)

- **PO1. Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- **PO2. Problem Analysis:** Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- **PO3. Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations
- **PO4. Conduct Investigations of Complex Problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- **PO5. Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
- **PO6. The Engineer and Society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.



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Program Outcomes (Pos) ..

- **PO7. Environment and Sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development
- **PO8. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- **PO9. Individual and Team Work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- **PO10. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- **PO11. Project Management and Finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- **PO12. Life-long Learning:** Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest of technological change.



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Program Specific Outcome (PSOs)

- **PSO1.** Graduate will attain in depth knowledge and hands on training for different technologies of computer science and engineering as well as information technology.
- **PSO2.** Graduates will effectively exhibit their skills and knowledge as ethical computer engineer with environment sustainability.
- **PSO3.** Graduate will be able to create solutions for real world complex problems in area of computer science and engineering
- **PSO4.** Graduates will be able to create, compare and develop new hardware as well as software technologies using advanced modern tools and languages leading to green and eco-friendly computing.



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COMPILER DESIGN

A compiler is a computer program that helps in translating the computer code from one programming language into another language. Basically, it translates the program written in the source language to the machine language. The compiling process contains an essential translation operation and error detection.



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Course Outcomes

- **TCS-352.CO1:** Utilize the concept of automata in the process of designing a compiler.
- **TCS-352.CO2:** Contrast the various parser required for constructing a compiler.
- **TCS-352.CO3:** Design an effective intermediate and optimized code generator i.e., three address code.
- **TCS-352.CO4:** Inference the effective runtime environment required for a compiler.
- **TCS-352.CO5:** Analyze the various code optimization techniques to optimize the intermediate code.



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Syllabus of course

● UNIT I

- ❖ Basics of Automata, Context-free Grammar; Finite automata- Deterministic finite automata, Non-deterministic finite automata; Phases and passes of a compiler; Bootstrapping, cross compiler; Compiler writing tools, LEX and YACC; Tokens, Patterns, Lexemes, Input-buffering.

● UNIT II

- ❖ Introduction to Parsing, Top down parsing, Problems associated with top down parsing; FIRST and FOLLOW; construction of LL(1) parsing table; Operator precedence parsing; Bottom up parsing, shift reduce parsing; Construction of LR[0] set of items, Construction of set of LR[1] items; Creation of SLR(1), CLR(1) and LALR(1) parsing tables, Parser conflicts; Handling of ambiguous grammar.



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Syllabus of course

● UNIT- III

- ❖ Introduction to syntax-directed Translation scheme; Intermediate code, postfix notation, Parse trees & syntax trees; Three address codes, Quadruple, Triples, Indirect Triples; Three address code for conditional statements, loops, arrays and case statements.

● UNIT-IV

- ❖ Data structure for symbols tables; Representing scope information; Run-Time Administration, Implementation of simple stack allocation scheme.
- ❖ Error Detection & Recovery, Lexical Phase errors, syntactic phase errors semantic errors.

● UNIT-V

- ❖ Loop optimization, the DAG representation of basic blocks, Creation of basic blocks, flow graph; Elimination of loop invariant computations, Elimination of induction variable, Reduction in strength, Loop jamming, Loop unrolling; Peephole optimization, common sub-expression elimination.



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Assessment Process (Theory and Practical)

- Midterm- 20 Marks
- Assignment : 10 Marks
- Teacher Assessment: 10 Marks
- External: 60 Marks



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Attendance Requirement

- Attendance Required for appearing in examination: 75 %



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Books

Text Book

1. "Principles of Compiler Design", Alfred V. Aho, Jeffrey D. Ullman, Narosa publishing house

● ***Reference Books:***

1. **Compiler Design; Chattopadhyay Santanu; Phi Learning [2009]**
2. **"Compilers, principles, techniques and tools" Alfred V. Aho, Ravi Sethi Pearson education.**



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Feedback System

- Feedback enables a teacher to see how teaching practice can be improved, and which teaching and learning strategies are more likely to be effective.
- At Uttaranchal University Daily feedback is given by students on per day lectures also, faculty and student feedback is given in End of the semesters in order to make changes in curriculum/courses(if required).



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Important Instructions Do and Don't

- Be regular in attendance and never be late
- Turn off your cell phone. Turn it completely off, not just to vibrate.
- Finish your assignments and read ahead. ...
- Avoid food distractions. ...
- Don't sleep in class. ...
- Don't engage in private conversation or activity. ...
- Don't show up late or leave early. ...
- Don't surf.



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Thank you!